



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

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MongoDB Atlas: Setup, Access and Cloud Management

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MongoDB Atlas – Introduction

► What is MongoDB Atlas?

MongoDB Atlas is a fully managed cloud database service
Provided officially by MongoDB Inc.
Removes the need for manual database administration

► Why Atlas is important

Simplifies deployment and scaling of MongoDB
Built-in security, monitoring and backups
Used widely in production cloud applications



► Cloud providers supported

Amazon Web Services (AWS)

Microsoft Azure

Google Cloud Platform (GCP)

► Multi-cloud advantage

Freedom to choose cloud provider

Geographic flexibility for deployments

Vendor lock-in is avoided



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Atlas Architecture – High Level View

► **Cluster-based architecture**

Atlas databases run inside clusters

Each cluster contains replica sets

High availability is enabled by default

► **Managed components**

Automated backups

Monitoring and alerts

Automatic patching and upgrades



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Creating a MongoDB Atlas Account

▶ Account creation

Sign up using email or GitHub account

No credit card required for free tier

Web-based dashboard provided

▶ Project concept

All resources are grouped under projects

Projects manage users, clusters and security

One account can have multiple projects



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Cluster Setup in MongoDB Atlas

► Cluster creation

Choose cluster type (Free, Shared, Dedicated)

Select cloud provider and region

Default replica set is automatically created

► Free tier clusters

Suitable for learning and development

Limited storage and performance

Ideal for student experiments



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Access Control and Security in Atlas

► Database users

Users are created inside Atlas

Roles define read and write permissions

Follows principle of least privilege

► Network access

IP whitelist controls who can connect

Prevents unauthorised access

Can allow specific IPs or cloud services



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Connecting Applications to Atlas

► Connection string

Atlas provides a ready-made connection URI
Includes cluster address and authentication
Used by applications and shells

► Supported clients

MongoDB Shell (mongosh)
Programming languages (Python, Java, Node.js)
GUI tools such as MongoDB Compass



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Monitoring and Performance Management

► **Built-in monitoring**

- CPU, memory and disk usage metrics
- Query performance statistics
- Real-time dashboards

► **Alerts and notifications**

- Alerts for high resource usage
- Email and webhook notifications
- Helps detect issues early



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Backups and Maintenance in Atlas

▶ **Automated backups**

- Continuous and snapshot backups supported
- Point-in-time recovery available
- No manual intervention required

▶ **Maintenance tasks**

- Automatic MongoDB version upgrades
- Security patches applied by Atlas
- Minimal downtime during maintenance



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Summary – MongoDB Atlas

► Key learning points

Atlas is a fully managed MongoDB cloud service
Provides security, monitoring and backups
Simplifies deployment and scaling

► Exam-focused keywords

Atlas, cluster, project, replica set
IP whitelist, roles, connection string
Monitoring, backup, cloud management



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UNIVERSITY

CELEBRATING 50 YEARS

Pooja T S
Assistant Professor
Department of Computer Applications
poojats@pes.edu
080-26721983 Extn: 391